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Democratizing artificial intelligence: How no-code AI can leverage machine learning operations

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Abstract Organizations are increasingly seeking to generate value and insights from their data by integrating advances in artificial intelligence (AI) (e.g., machine learning (ML) systems) into their operations. However, there are several managerial challenges associated with ML operations (MLOps). In this article, we outline three key challenges and discuss how an emerging type of AI platform—no-code AI—may help organizations address and overcome them. We outline how no-code AI can leverage MLOps by closing the gap between business and technology experts, enabling faster iterations between problems and solutions, and aiding infrastructure management. After outlining the important remaining challenges associated with no-code AI and MLOps, we propose three managerial recommendations. By doing so, we provide insights into an important emerging phenomenon in AI software and set the stage for further research in the area.
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1. The next generation of AI platforms

The use of software development tools that require little or no coding skills—referred to as *no-code software*—has been reported in prior literature. However, an emerging form of no-code tools that enable individuals to develop artificial intelligence (AI) solutions has gained less attention from scholars, despite forecasts of their rapid uptake through 2023 (Gartner, 2022; Marr, 2022). Here, we address part of this research gap by outlining how an emerging form of user-friendly tools for developing artificial intelligence (i.e., *no-code AI*) can support machine learning operations (MLOps) and help organizations derive value from intelligent systems.

The last decade has observed substantial advances in AI, including rapid developments in

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machine learning (ML; Jordan & Mitchell, 2015) and deep learning (DL; LeCun et al., 2015). Massive increases in the processing power of digital technology and available data have set the stage for increases in its use within numerous contexts (Dwivedi et al., 2021). As such, organizations are increasingly deploying AI technologies that can parse through large amounts of data, acquire skills and knowledge, and operate autonomously (Simsek et al., 2019). *ML* has been defined as "a broad subset of artificial intelligence, wherein a computer program applies algorithms and statistical models to construct complex patterns of inference within data" (Leavitt et al., 2021, p. 750. In addition, ML has proven to be particularly useful in organizational contexts (Sturm et al., 2021). To leverage value from it, many organizations are engaging in systematic ways of developing, deploying, and monitoring MLOps, as described in Section 2.

In addition, a new generation of digital platforms of increasingly human-AI hybrid systems is emerging (Rai et al., 2019). Here, we focus on part of this next generation (i.e., AI as a Service (AIaaS) platforms, or AI service platforms; Geske et al., 2021; Lins et al., 2021). These platforms have only recently gained researchers' attention, so their conceptualization has been limited as the literature is emerging and fragmented. We refer to this new generation of AI systems as no-code since common features are easy-to-use graphical user interfaces, which help novice users train ML models. However, they may also be included in larger software ensembles that provide more coding-intense approaches, or hybrid (low-code) modes. In the literature, these platforms are described as "cloud-based systems providing on-demand services to organizations and individuals to deploy, develop, train and manage AI models" (Lins et al., 2021, p. 442), and they are becoming more common—partly due to low adoption thresholds. Emerging evidence suggests that no-code or low-code tools can support ML pipelines (Li & Wu, 2022). The strategic importance of these tools has been recognized in previous research (Herremans, 2021) with additional measures that help organizations use AI in digital transformation initiatives (Holmström, 2022) by developing new capabilities and partnerships between AI and humans (Jarrahi et al., 2023). Typical features of no-code AI platforms (e.g., graphical interfaces) enable users to engage in training AI models without any coding and we argue that they have high potential utility in operationalization of ML. Thus, their current and potential roles in MLOps warrant detailed attention.

No-code AI platforms can make AI accessible and affordable for organizations that lack the size and technological competence needed for solutions that require advanced coding skills. In addition, they also guide users through the process of developing and deploying AI models without needing to learn the detailed workings of complex algorithms. Moreover, since the platforms are often cloud-based, no-code AI platforms remove concerns regarding installation, maintenance, and similar management problems. Large tech companies such as Amazon, Google, IBM, and Microsoft initially offered AI capabilities as a service, but smaller companies soon followed this trend, resulting in a plethora of available AI service solutions (AzureML Team, 2016; Bhattacharjee et al., 2017). As argued by Lins et al. (2021), the use of cloud technology in AI solutions enables organizations to access ubiquitous, convenient, and on-demand resources that may be rapidly provisioned. Against this backdrop, we address how no-code AI can support organizations' operationalization of ML by asking: How can no-code AI support MLOps?

To answer this question, we outline how AI tools can be adopted by diverse stakeholders. Combined with strategies that promote bottom-up approaches, this has the potential to enable the democratization (Von Hippel, 2005) of AI in organizational settings. By shifting the required capabilities for developing ML from a domain restricted to data scientists to a much broader domain, no-code AI can thus enable new paths for the development of intelligent systems.

The rest of the article is structured as follows. Section 2 focuses on the challenges experienced in MLOps today. Section 3 summarizes contemporary ideas about no-code AI. Section 4 addresses the RQ by exploring the intersection between MLOps and no-code AI, and Section 5 provides a brief conclusion of our discussion. Via these steps, this article contributes to an emerging stream of literature on no-code AI. By drawing upon knowledge of MLOps—combined with previous concepts from no-code software development—we highlight the benefits of no-code AI, provide specific suggestions for achieving them, discuss important remaining managerial challenges related to MLOps, and outline potentially fruitful avenues for future research.

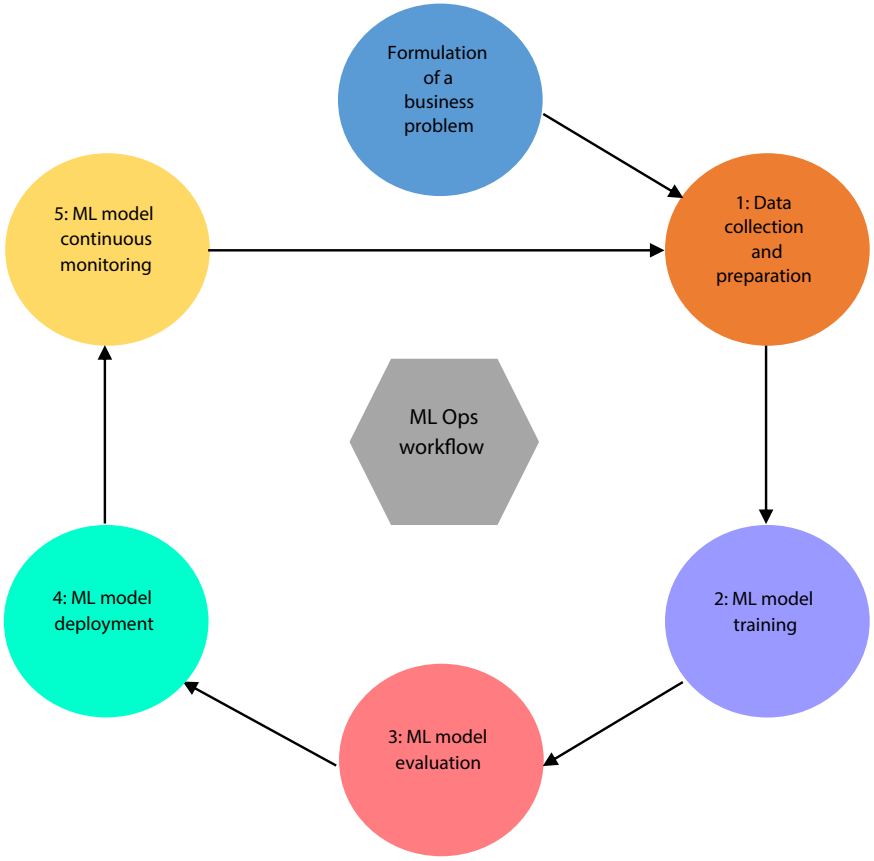
2. Challenges in machine learning operations (MLOPS)

The adoption of continuous software engineering practices such as development operations

(DevOps) in business operations has contributed to a significant shortening of software development and deployment cycles (John et al., 2021). The widespread use of ML in a corresponding approach (i.e., MLOps), as well as the role of no-code AI in the operationalization of ML, have received increased research attention (Tamburri, 2020). The development of ML significantly differs from traditional software development as ML models are dynamic and constantly changing (Borg, 2022; Moreschini et al., 2022). As such, there is a greater need for reliable production of ML-based applications and supporting infrastructure (Zhou et al., 2020). To develop responsible AI, it is vital to find appropriate tools and establish a culture of openness and learning (Matsui & Goya, 2022).

In this article, we apply a simplified version of the ML workflow as a template to discuss challenges linked to MLOps and the potential benefits of using no-code AI to address them. As illustrated in Figure 1, the ML workflow has five stages. The first is preparing collected data to train an AI model with an ML algorithm. Associated activities often include annotating data as most ML solutions used in organizations today are supervised and need labels to learn the content of an image or text, for example (Fredriksson et al., 2020). This step should not be underestimated, and data work accounts for most of the time spent on developing ML (Sambasivan et al., 2021). In the second (model development) stage, algorithms (e.g., decision trees and neural networks) and settings are tested and adjusted appropriately to create an ML model that performs well. Performance is evaluated using relevant measures (e.g., accuracy, precision, and recall) during stage three. The fourth stage starts when the model is regarded as providing sufficient performance (e.g., by correctly classifying a majority of cases). In this stage, it is integrated into and made accessible (deployed) to relevant resources in the host organization for generating predictions from data that it has not previously encountered. The fifth stage includes efforts to ensure that the model is current, establish measures to continuously monitor its performance and, when necessary, update it. As illustrated by the arrow between stage 5 and stage 1 in Figure 1, deployment is not the final stage of

Figure 1. The MLOps workflow



the ML workflow. ML models are dynamic and sensitive to changes in the environment in which they are used. Thus, they tend to degrade over time unless these changes are considered and accounted for using new datasets (Constantiou & Kallinikos, 2015) and additional cycles of training and evaluation.

However, only a small percentage of ML projects reach the production stage. MLOps is a response to the challenges associated with incorporating ML models into production. It is analogous to DevOps, which refers to a set of practices, tools, and a cultural philosophy that automates and integrates the processes involved in traditional software development from inception to retirement. As development teams become larger, several features of MLOps become increasingly important, including using appropriate frameworks, languages, tools, evaluation standards, change management procedures, and both monitoring and updating deployed models. Its use has been explored in several domains, including medicine (Granlund, Stirbu et al., 2021), finance (Xu, 2022), electricity supply (Subramanya et al., 2022), industry 4.0 (Faubel et al., 2022) and multiorganizational settings (Granlund, Kopponen et al., 2021). Although MLOps is a step toward systematic uses of ML models (Symeonidis et al., 2022; Tamburri, 2020), MLOps practice and research are still in the early stages (Bowne-Anderson, 2022), and many challenges remain. We outline three MLOps challenges throughout Section 2.

2.1. Challenge 1: Gaps between domain experts and technical experts

As noted by Garg et al. (2021), the multiple sets of actors involved in MLOps are often not synchronized and interact relatively late in the solution design process. Common problems are communication gaps between AI experts and domain experts. In other words, they do not speak the same language, and there are differences in both their knowledge and objectives. AI experts tend to talk in technical terms about ML models and how to optimize them, while domain experts are interested in solving problems related to their organizational responsibilities. In the long run, this may reduce the organization's ability to generate meaningful datasets or even reinforce biases due to a lack of sensitivity to the operating context. As such, there is a need for close collaboration between technical experts (i.e., data scientists and engineers) and domain experts that know the core businesses of the organizations (Testi et al., 2022; van den Broek et al., 2021). While this applies to IT and software development generally, it is even more important in the context of AI and ML as the development and use of these systems are heavily associated with professional cultures within computer and data science (Dourish, 2016).

2.2. Challenge 2: Slow problem-solution iterations due to ineffective architectures

ML originates from an academic tradition involving the use of nonnative software packages and academic datasets. However, such solutions do not scale well (Gift & Deza, 2021) and may be difficult to transfer into business settings, as they are primarily developed for research rather than a business environment. They are also largely developed by data scientists in experimental settings aimed at creating proofs of concepts and optimizing algorithms rather than creating solutions that can be deployed in organizational environments. Moreover, real-world settings are usually messier than controlled lab settings, and models based on lab work need to be tested in the environments where they are supposed to operate. While established platform enterprises that were born digital have already established both ML infrastructures and MLOps, these structures are more challenging to set up for traditional firms. Unfortunately, due to the scarcity of data science expertise, nonautomated data pipelines, and manual approaches to model training and comparison, iterations between problems and solutions are generally slow.

2.3. Challenge 3: Infrastructure management

As organizations progress in building and deploying ML, the demand for infrastructure that can handle these processes also grows. This progress requires efficient pipelines for data manipulation, model creation, and deployment (John et al., 2021; Mäkinen et al., 2021; Symeonidis et al., 2022). ML model training also depends on the presence of computational hardware such as graphical processing units (GPUs). Such hardware was once created to enable 3D graphics in computer games and is now at the core of the ML workflow as it can accelerate computationally demanding processes, such as deep learning (Pandey et al., 2022). Moreover, the infrastructure must provide the resources required to train new ML models amid changes in the organizational environment and requirements for new datasets. Deployed models must also be properly monitored, and systems must be established to ensure they meet key

criteria (e.g., sustainability, robustness, fairness, and explainability; Tamburri, 2020).

3. The emergence of no-code AI

No-code and low-code solutions for software development have been addressed in previous research as they enable nonprogrammers with little or no coding experience to produce various applications (Bhattacharyya & Kumar, 2023; Lethbridge, 2021; Luo et al., 2021; Richardson et al., 2014; Sahay et al., 2020; Yan, 2021). By adopting such solutions, enterprises may not only save time and costs but also see increases in development pace and product quality as the gap between business and information technology shrinks (Rokis & Kirikova, 2022).

As noted by Sundberg and Holmström (2022), a new breed of lightweight AI platforms enables nondata scientists to train ML models to make predictions. Examples of such platforms include BigML, Google AutoML, Microsoft Azure ML, Clarifai, and Levity (see Pawar et al., 2021). These platforms, also known as AI as a service (Lins et al., 2021) or simply AI service platforms (Geske et al., 2021), may keep up with or even outperform coded solutions (Kling et al., 2022). Hence, no-code AI platforms may be widely applied in various settings, including citizen science, as low-cost solutions in emerging markets, and by small and medium enterprises (SMEs). In the long run, it has been argued that the availability of user-friendly, low-code AI could democratize these systems' adoption and stimulate their multidisciplinary use (How et al., 2021). New drag-and-drop interfaces enable anyone to develop, train, and test AI algorithms in just a few hours. Thus, model development and deployment can be simplified, and—combined with a range of open-source solutions and plugins—organizational AI development can be flexible and context-sensitive, as discussed for example by Goffin (2021).

Koppe and Schatz (2021) highlighted the potential cost-effectiveness of cloud-based ML services, which can simplify the use of AI technologies by supplying pretrained neural networks as part of cloud services. In addition, they often support the whole development process, from uploading data to model deployment, even with smaller datasets than those traditionally required due to the use of transfer learning (e.g., via pretrained models such as the Bidirectional Encoder Representations from Transformers or BERT) and models based on ImageNet. Elshawi et al. (2018) referred to a range of cloud-based ML services, such as IBM Watson and BigML. Pawar et al. (2021) described *ML services* as integrated and semiautomated web devices that can aid users during the ML workflow and bridge the outcomes of these efforts with organizations' inner IT infrastructure via REST application programming interfaces (APIs).

An emerging trend can be found in the development of AI platforms as general-use products, with diverse potential applications and user-friendly, lightweight interfaces. In this context, scholars describe a need for AI platforms that are easy to use in several operational contexts and lower the threshold for inexperienced users. For example, Gao et al. (2021) argued that ML and DL approaches are valuable in the development of drugs for medical uses, but the lack of code-free and user-friendly applications prevents the use of such tools by domain experts.

The key advantage of no-code platforms is that they enable developers with different skill levels to create and deploy applications via graphical drag-and-drop interfaces to seamlessly connect all steps involved in the process. This enables rapid tests of ideas and faster initiation of projects and/or business ventures. As no-code AI tools have become mainstream, they have attracted the attention of organizations of all sizes, including startups and incumbent companies. No-code AI tools provide a user experience (UX) application layer combined with a purpose-built data science layer, which enables the cost-effective development of operational AI for many companies, as described by Geske et al. (2021).

4. How no-code AI can leverage machine learning operations

In Section 2, we presented three challenges associated with MLOps today: (1) gaps between business and information technology experts, (2) slow production of AI models, and (3) infrastructural demands associated with higher complexity. In the following subsections, we outline how no-code AI may address these challenges.

4.1. Closing the gap between business and technology experts

While no-code AI provides solutions without writing code, it also enables developers to focus on writing important code by automating certain points of the ML workflow and spreading the responsibility for and engagement in these flows. To

do this, developers utilize more of the host organization, turning no-coders into developers (Yan, 2021). This reduces the impact of skill shortages and unifies organizational competencies. Instead of focusing on coding, teams can prioritize critical business goals and data work, thereby developing use cases and practical applications of ML. In the long run, this establishes important cross-skills, which reduces the reliance on a few individuals in the organization. As employees with different competencies and skills engage in the ML workflow, the gaps between domain experts and technical experts (e.g., data scientists) shrink.

Thus, no-code AI can be an important facilitator that promotes interactions between actors who previously would not—or could not—engage with each other. It can potentially enable more actors to access AI and promote its democratization. As a range of no-code solutions becomes commonplace, users may train ML models with very little or no prior coding experience. As such, these new platforms have been referred to as *lightweight AI*, as they bridge gaps between data experts and more novice users and enable access for broader sets of users (e.g., nonexperts that interact with intelligent systems), with much less need for heavy investment in new competencies (Sundberg & Holmström, 2022).

When businesses aim to develop operational AI and ML systems, it is crucial to involve domain experts in the developmental efforts—and no-code solutions grant access to tools for developing them without coding experience. This is crucial for establishing relevant ground truths based on rich knowledge and the selection of appropriate means for transfer learning. Activities such as data annotation, augmentation, and interpretation of the output of ML models are all highly dependent on the knowledge of domain experts (Lebovitz et al., 2021; van den Broek et al., 2021). However, van den Broek et al. (2021) referred to the notion of *hybrid practices*, in which ML experts engage with domain experts during specific stages in the development of predictive algorithms. By contrast, we highlight how these interactions can be made continuous and permanent via comprehensive solutions that can be understood and used by domain experts. One example involves simplifying the data annotation process via intuitive interfaces in which domain experts can label text, images, or objects in images based on their knowledge and experience. This can promote the establishment of cross-functional teams in an organization, with the core technologies stimulating the generation of a common language regarding AI. Where previous research has assigned domain experts a minor role in the MLOps workflow,[1] we expect their engagement to deepen as AI development continues.

The inclusion of domain-wide expertise in the ML workflow is also important for organizations' attempts to diversify by creating and making sense of their data (e.g., via visualizations). Here, we see an interesting contrast between the use of shared datasets and pretrained models to optimize performance with the adaptation of ML to a firm's local conditions. Thus, by adopting no-code AI, firms can develop ambidextrous ML systems that provide both sensitivity to local conditions and robust communication with global AI ecosystems. This can be done by establishing communities that use the platforms to create and share models, scripts, and solutions that can be deployed by other users in the ecosystem. For example, the community resources established around the Hugging Face platform include forums and collaborative spaces, as well as a large range of pretrained models and opportunities to export and import solutions to other systems. Moreover, there is also emerging evidence that no-code solutions can aid the operationalization of large generative language models (transformers) such as the highly popular Generative Pretrained Transformer 4 (GPT-4; Kim et al., 2021). Recently, Microsoft has integrated several such models in their Azure OpenAI service as a part of its partnership with OpenAI. Amazon is on a similar path to make a vast range of pretrained models available in its SageMaker platform, via a collaboration with Hugging Face. These efforts enable the adaptation and integration of popular models made for different purposes (e.g., ChatGPT, Codex, and DALL·E 2) in various settings. As generative AI models are becoming more integrated with different kinds of systems, we expect them to be a part of many AI solutions in the near future.

4.2. Enabling faster iterations between problems and solutions

One of the few empirical investigations of no-code AI suggests that the examined platform "enables quick prototyping, rapid experimentation, and reduces the time to production by standardizing model building and deployment" (Iyer et al., 2021, p. 33). By democratizing access to novel ML technologies—such as DL algorithms—no-code AI allows organizations to develop and deploy their own AI solutions quickly by providing prototyping

[1] See, for example, Ruf et al. (2021)

capabilities (Kim et al., 2021; Li & Wu, 2022). However, a related problem is the scarcity of available AI expertise. The use of no-code AI reduces the need for AI experts—or, rather, it enables them to focus on their core tasks by automating parts of the ML workflow, which diminishes the common gaps between AI experts and domain experts. No-code AI can help AI experts meet optimization objectives and enable domain experts to test their ideas and generate value for their firms without needing to rely on coders (see Bhattacharyya & Kumar, 2023). For example, with the click of a button, the OptiML function of the BigML platform tests hundreds of different models based on common algorithms (e.g., via decision trees, neural networks, and automatic comparison of performance).

With appropriate management, these features enable the establishment of well-balanced and integrated MLOps processes throughout an organization. This greatly enhances the speed and efficiency of ML model development, as well as communication and collaboration between teams with streamlined interorganizational relationships. In this manner, no-code solutions can level the playing field between SMEs and large organizations due to their cost-effectiveness (see Bhattacharyya & Kumar, 2023). They provide a high degree of freedom for diverse businesses to offer generic AI services and expand their scope of services (Geske et al., 2021) (i.e., boundaries; Holmström, 2022). Moreover, no-code and low-code tools accelerate the development of AI systems since they are ready to use and provide APIs and other out-of-the-box components (Luo et al., 2021). *Inter alia*, their use accelerates growth by helping businesses engage in rapid prototype development (Yan, 2021). This enables rapid time-to-market and deployment of AI services, increased performance and resilience, and lower dependency on AI engineers who may be challenging to recruit (Lins et al., 2021).

By enabling the rapid integration and deployment of ML solutions and their large-scale adoption, no-code AI reduces the risks of AI solutions ending up on the ever-growing proof of concept (POC)-scrap pile. Performance enhancements via customization and optimization of AI models lead to improvements in flexibility and cost/benefit ratios (Lins et al., 2021). No-code AI provides important foundations for robust interfirm AI ecosystems. In addition to enabling cloud-supported scaling capabilities and helping build and deploy applications (Sahay et al., 2020), these platforms also provide access to computing resources and expertise, automatic selection (e.g., of classifier models; Lins et al., 2021), and engagement within networks and communities.

4.3. Aiding infrastructure management

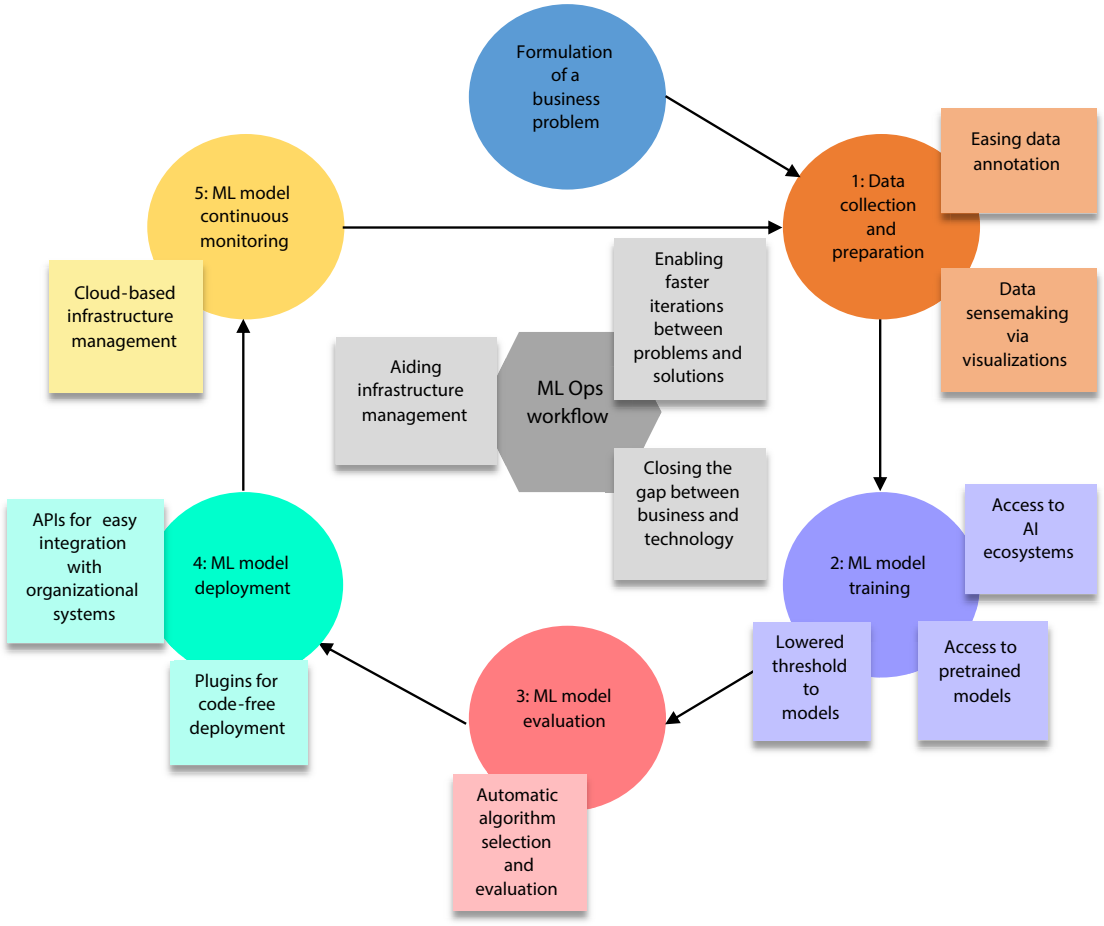
As mentioned, when organizations progress in the building and deployment of ML systems, the infrastructure required to handle the associated processes increases. Common components include data pipelines, training setups, evaluation measures, and postdeployment performance assessment routines for continuous monitoring and training. Without appropriate infrastructure, ML systems may become hazardous and irresponsibly deployed rather than well-controlled and continuously monitored. By using no-code AI, organizations can focus on core competencies and businesses supported by cloud AI services (Lins et al., 2021), as well as engage in the responsible development of AI. It can also increase security by delegating maintenance to service providers (Yan, 2021), which reduces the need to troubleshoot cross-tool functions and related infrastructure. The AI service provider manages hardware and software infrastructure, so users do not need to handle system setup and configuration (Lins et al., 2021). Moreover, ML services involve integrated and semiautomated functions for preprocessing, designing, and deploying models with an organization's IT infrastructure via APIs. Several no-code solutions—such as Bubble—are also available to build plugins for calling ML models via these APIs, to create apps for generating predictions.

ML infrastructure includes the tools and resources that constitute the AI environment. A well-managed infrastructure accurately tracks the causes of failure and facilitates corrections, as no-code AI can streamline important components of the development environment. Moreover, no-code solutions support interaction with external services and data sources, increased security measures (e.g., authentication, security protocols, and user access control), and collaborative development and reuse of resources (Sahay et al., 2020). No-code AI even facilitates access to additional resources such as pretrained ML models (Geske et al., 2021), and cloud services enable parallelization via cost savings due to flexible cost models (Lins et al., 2021).

4.4. Key managerial challenges for no-code AI

As summarized in Figure 2, no-code AI tools have many benefits for organizations. However, they do not come without substantial challenges, as outlined in Section 4.4.

Figure 2. How no-code AI may support MLOps



4.4.1. No-code AI requires strategy

As mentioned by Bhattacharyya and Kumar (2023), no-code solutions must be accompanied by robust adoption strategies—otherwise, employees may have low perceptions of utility. On a similar note, Luo et al. (2021) reported conflicting views of the advantages and disadvantages of no-code solutions among practitioners, which may be associated with trade-offs between ease of use and user control (Lins et al., 2021). Simply providing easy access to tools is not enough to democratize novel technologies. Rather, these efforts must be accompanied by participatory practices that inspire a large range of stakeholders to engage in bottom-up approaches (Björgvinsson et al., 2010).

4.4.2. Interoperability and scalability must be considered

To effectively integrate no-code AI services in organizational settings, sufficient interoperability and sound data governance and protection must be established and maintained (Lins et al., 2021). There are also risks for vendor lock-ins, and trust in the service provider is required for the use of no-code AI. In addition, no-code solutions are limited when highly customized solutions are required. While they can be scaled, they are rarely used for large-scale business applications (Yan, 2021). Organizations that already have traditional ML systems in place need to find paths that allow harmonious use of these systems with no-code solutions to avoid stove piping in their MLOps.

4.4.3. No-code AI cannot fix poor or biased data

Finally, while no-code AI facilitates data annotation via graphical interfaces, challenges related to data work persist (Mäkinen et al., 2021), and neither MLOps nor no-code AI is a panacea for poor or biased data. ML models decay rapidly with changes in the environment in which they are deployed. However, by involving actors with diverse competence in the ML workflow, we argue that the systematic use of no-code AI may lead to higher-quality datasets as more people are involved in their creation, labeling, and evaluation. This process is the most time-consuming part

of the ML workflow, and by freeing up resources in other stages of the flow, organizations can engage more people to ensure the data is high quality. Either way, no-code AI will not solve problems with biases in datasets nor remedy data quality problems. However, making intelligent techniques accessible to a larger range of users may pave the way for more responsible uses and more relevant data.

5. Concluding discussion and outlook for further research

This article explores an emerging form of AI platforms (i.e., no-code AI) by addressing one essential research question: How can no-code AI support Machine Learning Operations (MLOps)? Departing from challenges experienced in MLOps, we have outlined how no-code AI can leverage operational AI by closing gaps between business and information experts, enabling faster iterations between problems and solutions, and aiding infrastructure management.

AI is a key driver of digital transformation in today's organizations—largely due to its proficiency in decision-making when combined with large datasets. However, implementing AI can be a complex process that requires a clear understanding of the business goals and objectives, as well as the specific challenges the organization deploying it is facing. One way to address these challenges is by using no-code AI, which can help democratize AI by making it accessible to more people. No-code AI platforms allow individuals—regardless of their technical background—to build and deploy AI models with minimal coding. This can help increase the number of people who can participate in the AI deployment process, thereby diversifying perspectives and ideas. No-code AI platforms can also help organizations save time and resources as they do not need expensive data scientists or developers to build AI models. By engaging with no-code AI, organizations can develop important skills and relations mentioned in previous research (Jarrahi et al., 2023), such as reinventing and elevating others, developing AI literacy, facilitating mutual learning, and forming cross-functional teams.

Currently, a wave of no-code AI tools is becoming available for the masses and will contribute to AI's democratization and accessibility by simplifying the ML process. For future workers, no-code AI will be more than just a set of tools; it will reflect a culture and mindset. Our analysis—paired with previous studies—indicates that the following recommendations for managers may facilitate the effective adoption and deployment of no-code AI, as well development of an appropriate culture.

First, organizations must integrate no-code tools into their AI strategies (Herremans, 2021). As mentioned throughout this article, no-code AI can close the gap between business and technology experts, enable faster iterations between problems and solutions, and aid infrastructure management. By understanding these benefits, managers can better leverage no-code AI to generate value and insights from their data. Moreover, with the recent advances in generative AI, large language models will likely become integrated features in no-code platforms. By exploring these systems, managers can acquire the necessary tools and capabilities to implement ML models and workflows without requiring deep technical expertise. Investment in these platforms can help managers overcome the challenges associated with MLOps. As many platform providers offer free or low-cost options that support the full ML workflow, there are interesting paths for creating no-code AI champions (see Jarrahi et al., 2023) in an organization who can act as important facilitators and engage with additional employees—with certain limitations (e.g., dataset size and permitted number of trained models). This also enables periods of testing different types of platforms to identify which is most suitable for a particular organization. Then, organizations can proceed with suitable scaling activities and create collaborative spaces in the paid versions of their chosen platforms.

Second, managers should seek collaboration with technical experts to align business goals and data and model development. No-code AI platforms can help managers work more closely with technical experts, such as data scientists and engineers. Managers should look for opportunities to collaborate with these experts to better understand the underlying ML models and workflows and identify potential issues that may arise. Despite the benefits of no-code AI, there are still challenges that must be addressed, including security, scalability, and interpretability issues (Lins et al., 2021; Yan, 2021). As such, these tools will likely serve as part of a wider AI ecosystem in which they exist with more traditional ML development systems. Managers should understand the challenges of integrating these systems and take steps to maximize the potential success of no-code AI initiatives.

Third, managers should use no-code AI to achieve responsible AI. Through the creation of diverse, cross-functional teams, many of the issues

associated with the use of intelligent and predictive systems in organizations (e.g., data bias and irrelevant or harmful predictions) can be detected, properly managed, and mitigated. No-code AI platforms can enable diversity in data work and empower employees to monitor and evaluate the models' performance in near real-time, which allows them to quickly identify and address any issues. As domain experts are participating in the ML development workflow, changes in the environment that affect the models' performance can be identified and addressed more rapidly. Continuously monitoring and evaluating the performance of ML models can help managers ensure that they are providing accurate and actionable insights while checking that the outcomes align with best practices for developing responsible AI (Matsui & Goya, 2022).

In this article, we have sought to scrutinize and enhance the understanding of a novel, emergent phenomenon: no-code AI. We have summarized the associated benefits and challenges in diverse, fragmented streams of literature on no-code software development (i.e., MLOps) and AI as a service. We have also formulated the prevailing insights into three specific managerial recommendations, thereby paving paths for further analyses. Our proposals are in line with emergent research on MLOps that highlights the importance of openness and creating a learning organizational environment, as well as choosing the right tools for developing responsible AI (Matsui & Goya, 2022). We see several opportunities for scholars to engage in rich case studies of the future development and implementation of no-code AI in diverse organizational settings. Doing so will further illuminate the benefits and challenges we have presented here.

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